No.

5 If a and b in $\mathbb{Z}_n$ have the same additive inverse l, then

$$a = a + 0 = a + (l + b) = a + l + b = 0 + b = b$$

So different integers have different inverses

Thus $\{-0, -1, \ldots, -(n-1)\}$ is a set of n distinct elements in the n element set $\{0, 1, \ldots n-1\}$. So they are the same set

16 Answer: $15^{-1} = -3 = 43$

GCD alg

A	B
46	15
1	15

$46 + (-3)15 = 1$ which

Done.

21 The complementary design has blocks

$\{2, 4, 5, 6\}$, $\{0, 3, 5, 6\}$, $\{0, 1, 4, 6\}$
$\{0, 1, 2, 5\}$, $\{1, 2, 3, 6\}$, $\{0, 2, 3, 4\}$
$\{1, 3, 4, 5\}$

29 $\{0, 2, 5, 11\}$ isn't a difference set in $\mathbb{Z}_{12}$ because $5 - 2 = 3$ and $2 - 11 = 3$.

50 Let $x_1, x_2 \in \{0, \dots, m-1\}$ and $y_1, y_2 \in \{0, 1, \dots, n-1\}$. We find i and j such that

$$(A_1 \otimes B_1)_{ij} = (x_1, y_1) \quad \text{and} \quad (A_2 \otimes B_2)_{ij} = (x_2, y_2)$$

As there are $(mn)^2$ such choices, and each pair pair occurs, each occurs exactly once. So this will prove that $A_1 \otimes B_1$ and $A_2 \otimes B_2$ are orthogonal.

Let x_1, x_2, y_1, y_2 be chosen. As A_1 and A_2 are MOLS, there exist $\alpha, a \in \{0, \dots, m-1\}$ such that

$$(A_1)_{\alpha, a} = x_1 \quad \text{and} \quad (A_2)_{\alpha, a} = x_2.$$

Similarly, there are $\beta, b \in \{0, 1, \dots, n-1\}$ such that

$$(B_1)_{\beta, b} = y_1 \qquad (B_2)_{\beta, b} = y_2$$

So

$$(A_1 \otimes B_1)_{\alpha m + \beta, am + b} = (x_1, y_1)$$

and

$$(A_2 \otimes B_2)_{\alpha m + \beta, am + b} = (x_2, y_2)$$